

invertebrates. Leeches range in length from 1 to 30 cm. Most leeches inhabit fresh water, but there are also marine species and terrestrial leeches, which live in moist vegetation. Some parasitic species use bladelike jaws to slit the skin of their host. The host is usually oblivious to this attack because the leech secretes an anesthetic. After making the incision, the leech secretes a chemical, hirudin, which keeps the blood of the host from coagulating near the incision. The parasite then sucks as much blood as it can hold, often more than ten times its own weight. After this gorging, a leech can last for months without another meal.

Until the 20th century, leeches were frequently used for bloodletting. Today they are used to drain blood that accumulates in tissues following certain injuries or surgeries. In

addition, forms of hirudin produced with recombinant DNA techniques can be used to dissolve unwanted blood clots that form during surgery or as a result of heart disease.

Earthworms Earthworms eat their way through the soil, extracting nutrients as the soil passes through the alimentary canal. Undigested material, mixed with mucus secreted into the canal, is eliminated as fecal castings through the anus. Farmers value earthworms because the animals till and aerate the earth, and their castings improve the texture of the soil. (Charles Darwin estimated that one acre of farmland contains about 50,000 earthworms, producing 18 tons of castings per year.)

A guided tour of the anatomy of an earthworm, which is representative of annelids, is shown in **Figure 33.25**.

Each segment is surrounded by longitudinal muscle, which in turn is surrounded by circular muscle. Earthworms coordinate the contraction of these two sets of muscles to move.

Coelom. The coelom of the earthworm is partitioned by septa.

Metanephridium. Each segment of the worm contains a pair of excretory tubules, called metanephridia, that discharge wastes from the blood and coelomic fluid through exterior pores.

Many of the internal structures are repeated within each segment of the earthworm.

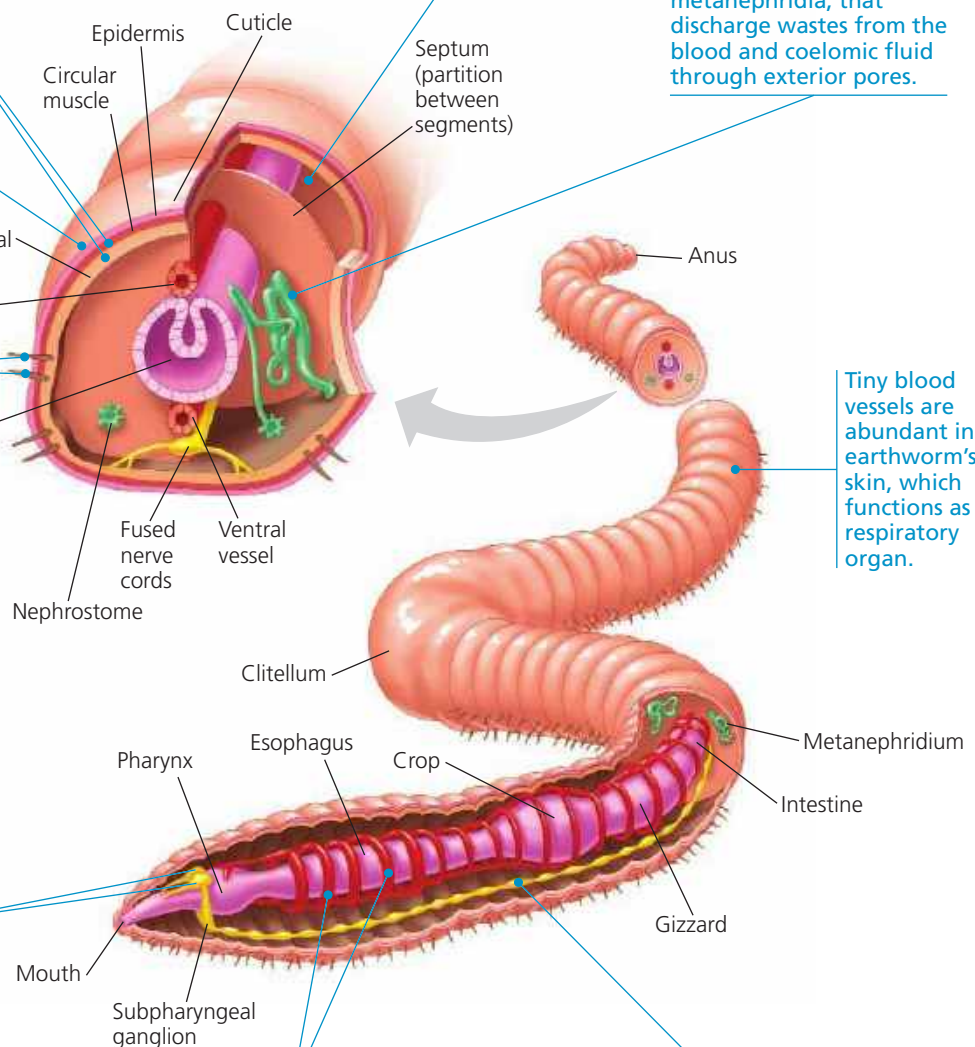
Chaetae. Each segment has four pairs of chaetae, bristles that provide traction for burrowing.



Giant Australian earthworm

Cerebral ganglia. The earthworm nervous system features a brain-like pair of cerebral ganglia above and in front of the pharynx. A ring of nerves around the pharynx connects to a subpharyngeal ganglion, from which a fused pair of nerve cords runs posteriorly.

▲ Figure 33.25 Anatomy of an earthworm, a sedentarian.



The circulatory system, a network of vessels, is closed. The dorsal and ventral vessels are linked by segmental pairs of vessels, some of which are muscular and pump blood through the circulatory system.

Ventral nerve cords. The nerve cords penetrate the septa and run the length of the animal, as do the digestive tract and longitudinal blood vessels.